KOHLER-MITWPU AI RESEARCH LAB PROGRAM · TRACK 2

KOHLER

AquaGuard AI

Prompts & AI Workflow Documentation



Commercial Smart Facility & Sustainability Manager

AI WORKFLOW

From Sensor Data to Action

01

Sense

Collect facility telemetry:

- Water flow
- Flush count
- Occupancy
- Leak sensor
- Sensor status
- Time since cleaning

02

Detect

Identify abnormal patterns using multiple signals.

03

Understand

Determine the likely cause and severity of the issue.

04

Predict

Estimate water loss, hygiene requirements and maintenance urgency.

05

Act

Generate a recommended maintenance or cleaning action.

CORE PRINCIPLE **Sense → Detect → Understand → Predict → Act**

MASTER SYSTEM INSTRUCTION

AquaGuard AI — System Instruction

ROLE

You are AquaGuard AI, an intelligent facility monitoring assistant.

Analyze restroom and facility telemetry and convert sensor information into clear, explainable and actionable insights.

CONSIDER

- Water flow
- Flush frequency
- Occupancy
- Leak sensor status
- Sensor health
- Time since cleaning

REASONING RULES

1. Do not rely on a single sensor when multiple signals are available.
2. Compare water flow with occupancy and flush activity.
3. Identify potential continuous water usage when flow remains high despite low activity.
4. Consider sensor warnings separately from water-related anomalies.
5. Identify cleaning requirements using usage and time since cleaning.
6. Explain why an alert was generated.
7. Recommend an appropriate action.
8. Clearly communicate uncertainty when the available data is insufficient.

OUTPUT GOAL **Detection → Explanation → Prediction → Recommended Action**

CORE AI PROMPTS

Four Prompts, One Workflow

Leak Detection Prompt

Analyze the facility telemetry and identify possible water leaks. Consider water flow, occupancy, flush count and leak sensor status together. Do not classify a restroom as a leak based only on one abnormal value. Provide the risk level, supporting signals and recommended action.

Predictive Hygiene Prompt

Analyze occupancy, flush activity and time since the last cleaning. Identify restrooms that may require cleaning based on actual usage rather than relying only on a fixed schedule. Explain the reason for the recommendation.

Sensor Health Prompt

Review the sensor status and identify devices requiring attention. Distinguish sensor-health problems from actual water or hygiene problems.

Maintenance Priority Prompt

Combine detected issues and classify maintenance actions according to urgency. Prioritize potential water loss and critical facility conditions while clearly explaining the reason for each priority.

B-203 AI REASONING EXAMPLE

Multi-Signal Leak Detection

Restroom B-203

Water Flow	18.8 L/min
Occupancy	20%
Flush Count	4
Leak Sensor	Triggered
Sensor Status	OK

AI REASONING

High water flow
18.8 L/min is significantly higher than normal facility readings.

Low occupancy
Only 20% occupancy is reported.

Low flush activity
Only 4 flushes are recorded.

Leak sensor triggered
An independent sensor also indicates a possible leak.

RISK LEVEL

CRITICAL

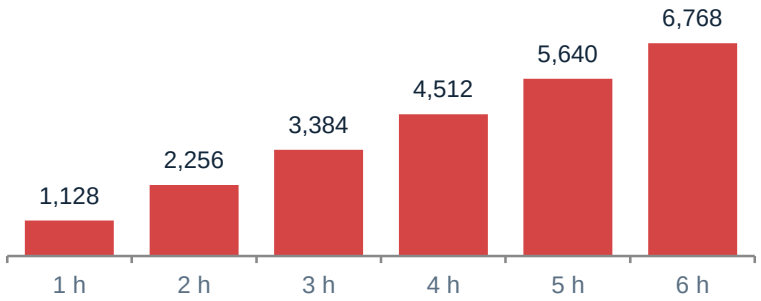
High-confidence potential continuous water leak from multiple independent signals.
Recommended action: Inspect B-203 immediately and verify the water source.

“What Happens If I Ignore It?”

B-203 · PROJECTED ADDITIONAL LOSS OVER 6 HOURS

6,768 L

at the current abnormal flow of 18.8 L/min, turning a technical alert into an understandable operational impact.



Ask AquaGuard

Facility teams can ask questions such as:

Why is B-203 critical?

AquaGuard
B-203 is critical because it has high water flow, low occupancy, low flush activity and a triggered leak sensor.

Alert → Explanation → Decision

Helps teams understand the reason behind an alert instead of simply displaying a warning.

STRUCTURED OUTPUT

Standard AI Decision Format

Facility / Restroom	Which location requires attention?
Issue	What abnormal condition was detected?
Evidence	Which sensor signals support the finding?
Risk Level	Healthy / Attention / Critical
Prediction	What could happen if the issue continues?
Recommended Action	What should the facility team do next?

EXAMPLE · B-203

LOCATION
B-203

ISSUE
Potential continuous water leak

EVIDENCE
High flow + low occupancy + low flush count + leak signal

RISK
Critical

IMPACT
6,768 L projected over 6 hours

ACTION
Immediate inspection

TRANSPARENCY & FUTURE SCOPE

Prototype Today, Production Tomorrow

PROTOTYPE TRANSPARENCY

The current system uses simulated IoT telemetry for demonstration. It does not use real KOHLER facility data. The prototype demonstrates the intelligence and workflow that could be connected to real facility infrastructure.

FUTURE PRODUCTION ARCHITECTURE



FUTURE INTEGRATIONS

- Live IoT sensor streams
- Enterprise maintenance systems
- Automated maintenance tickets
- Advanced ML anomaly detection
- Enterprise AI/LLM services
- Historical trend analysis

IMPACT

From Facility Data to Facility Intelligence

AquaGuard transforms raw facility telemetry into:



Explainable

Understand why an alert occurred.



Predictive

Identify potential problems before they become larger issues.



Actionable

Connect detection to maintenance and cleaning decisions.



Sustainable

Support earlier water-loss detection and more efficient facility operations.